

Synthesis and characterization of sustainable geopolymer green clay bricks: An alternative to burnt clay brick

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HIGHLIGHTS

- Synthesis of geopolymer green brick by using clay and fly ash as precursors.
- Optimize the proportion of both precursors to achieve the desired performance.
- Comparison of Physico-mechanical properties with the codes for burnt clay bricks.
- Characterization of geopolymer brick through FTIR and XRD analysis.
- Sustainable green brick was developed by using 40 wt% clay and 60 wt% fly ash.

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ABSTRACT

Clay is a material that is rich in oxides of aluminum and silicon, which are also abundant in fly ash, and have strong potential to be used as a precursor for the synthesis of a sustainable geopolymer green brick. This experimental work was designed to optimize the amount of clay and fly ash as precursors. Eleven mix proportions were designed, the clay was replaced by fly ash from 0 to 100 wt%, At macro-scale, the physical and mechanical properties of resultant geopolymer bricks were assessed and compared with the codes for a conventional fired clay brick. It was observed that the brick with 30 to 60 wt% fly ash having; low density, allowable water absorption and high compressive strength can be used as a load bearing material. At micro level, the geopolymerization was assessed by Fourier transform infrared (FTIR) spectroscopy, which was also verified by the anorthite peak in the X-ray diffraction (XRD) analysis. The XRD results are complimented with the FTIR and verify the gain in mechanical strength of geopolymer bricks by polycondensation. The resultant geopolymer green brick is a sustainable construction material with smooth manufacturing process along with superior physical and mechanical properties as compared to a conventional brick.

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1. Introduction

Brick is the oldest construction material known to humankind [1]. Over 1.3 trillion bricks are produced worldwide, annually. Pakistan, being 3rd largest brick producing country of South Asia, has 18,000 brick kilns that produce 45 billion bricks per annum [2]. Although bricks are being used for thousands of years, yet the irony

is the design and manufacturing process is hardly changed. The precursor is still the same; i.e. clay mixed with water. The practice can be categorized into seven simple steps, i.e. clay preparation, mixing, sun drying, packing into kiln, firing and then unpacking [3]. The traditional brick kiln is heated up to 1000–1500 °C, where, alumina, silica and iron are converted into fused glass which act like binding agent that provide adhesion. The kiln is then cooled for several days to normalize the temperature, so, the process is rather hectic and highly energy and time consuming [4].

The use of crushers for raw material extraction, mixers for mixing and relevant technology has helped a lot in making the process

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efficient yet it has certain disadvantages such as the machine baked bricks do not let cement stick to its face. On the other hand, the manual process is laborious, ineffective, time-taking, painstaking, less efficient, need large working space and adversely affecting the lives of kiln workers [5,6]. These restraints gave rise to the need for development of efficient methods and techniques for brick production.

Apart from methods of production the emerging concerns related sustainability led human beings towards investigations in order to formulate sustainable, lightweight and cheaper construction materials. In this regard, numerous researches have been performed to develop sustainable construction materials, by incorporating waste materials in bricks [6–9]. Moreover, the rapid wave of industrialization stimulated waste production which created a severe problem in terms of waste disposing and handling. Many researchers had investigated the use of by-products such as fly ash [7], slag [10] and silica fumes [11]. The success of these researches has amplified further research and has proved a great help for governments in terms of waste handling [6,12]. It has been noticed that the incorporation of waste in bricks reduces the compressive strength [13,14], so this area is of critical importance. The compressive strength of cigarette butt reinforced brick varies from 3 to 12 MPa which fulfils the minimum limit of 3 MPa for compressive strength of clay bricks as specified by Australian Standard AS/NZS 4455.1:2008 [15].

Fly ash is a by-product largely consists of the particles driven out after coal burning in coal fired power plants. The constituents may vary but the main components are silicon dioxide, aluminum oxide and calcium oxide. Minute percentages of chromium, arsenic and other metals are also found [16]. Fly ash is pozzolanic material which has siliceous and aluminous components that form cement [17]. It is classified into class C and class F, where class C fly ash contains higher amount of calcium [18]. Fly ash can be used as precursor as it can easily be activated and undergo enhanced geopolymerization reaction under certain curing conditions [19,20].

Geopolymers are covalently bonded non crystalline amorphous networks [21]. Solid aluminosilicates when comes in reaction with highly concentrated alkaline activator solutions produce alkali-aluminosilicate material, which is called geopolymer [22]. This material possesses properties like cementitious materials with a wide range of applications, considerably reducing the toxic emissions [23]. Mostly, owing to the precursor materials, the properties of resultant geopolymer may vary ranging from higher compressive strength, lower thermal conductivity and low or fast shrinking. Commonly known by the name of geopolymers, the material is also known as low-temperature synthesized aluminosilicates because they are manufactured at low temperature such as 100 °C [24].

A large number of studies have been carried out in the field of geopolymers. Researchers have studied geopolymers by using metakaolin and calcined clays [25]. It was found out that high liquid/ solid ratio, in case of geopolymerization of calcined clay, accelerated dissolution and hydrolysis. Another study used slag as precursor and the delay in setting time was investigated [26]. A similar research was carried out by using metakaolin and blast furnace slag as raw materials and concluded that geopolymers are excellent anti-corrosion materials, therefore, can be used in coating of marine concrete [27]. Up to authors knowledge not a single study has been conducted that investigated the raw clay as precursor in geopolymerization for synthesis of green brick. However, clay is rich in oxides of aluminum and silicon and have a strong potential to be used as precursor.

Therefore, the present study is carried out by making a sustainable construction materials called geopolymer green brick by using clay and fly ash as precursors along with alkaline activators. Both macro-scale and micro-scale properties were checked, the physical and mechanical properties of bricks were assessed at macro level,

whereas degree of geopolymerization was assessed at micro level. The optimum composition developed were 40 wt% fly ash 60 wt% clay, 50 wt% fly ash 50 wt% clay, and 60 wt% fly ash 40 wt% clay, 18 wt% Na₂SiO₃, 7 wt% NaOH and 2 wt% water.

1.1. Significance of research

As compared to the conventional methods this research utilizes locally available precursor materials with geopolymerization technique. This method leads to fuel economy as lesser amounts of energy is required for this method i.e. 70–110 °C curing temperature, consistently decreasing the amount of CO₂ emissions. As a result, a green sustainable construction material, with superior mechanical properties in terms of compressive strength, was developed. In addition, due to adaptation of oven curing method and geopolymerization process, a substantial amount of water is conserved in this geopolymer green brick production. Hence the production of geopolymer green bricks will be a smooth process without causing much fatigue and also providing a sustainable construction material with superior mechanical and physical characteristics.

2. Experimental methodology

2.1. Materials

The precursor materials used to synthesize geopolymer green brick were clay and fly ash, whereas sodium hydroxide (NaOH) and sodium silicate (Na₂SiO₃) were used as alkaline activators. The clay used in the investigation was collected from a local brick kiln in the Punjab province of the Pakistan in order to reach at a concrete conclusion when compared with fired clay brick properties. The characterization of clay was carried out according to ASTM D-4318 [28] and particle size was analyzed by sieve analysis according to ASTM D-6913 [29]. Physical characteristics of the clay are described in Table 1. The fly ash was collected from DG cement Punjab Pakistan and its chemical composition is provided in Table 2. It can be observed that fly ash falls in the category of Class F according to ASTM C-618 [30]. It was observed that minerals like quartz, mullite, anorthite, calcite, hematite and magnetite are present in fly ash. The NaOH solution of 14 Molar along with Na₂SiO₃ (water glass) of specific gravity 1.35 g/cm³ were used. The densities of precursor materials and alkaline activators were calculated and presented in Table 3.

2.2. Specimen preparation

Geopolymer green brick was cast by mixing the precursors and alkaline activators. The amount of solids was 73 wt% and liquid was 27 wt%. The solid constituents were clay and fly ash and their proportions were varied to cast 11 types of specimens; clay was replaced by fly ash by an amount of 0–100% by weight. The details of mix proportions for each type of specimen are mentioned in

Table 1
Physical properties of clay.

Description	Value
Specific gravity	2.75
Liquid limit (%)	32
Plastic limit (%)	26
Grain size distribution analysis	
Gravel %	0
Sand %	3
Silt & Clay %	97

Table 2
Chemical proportion of fly ash.

Description	Oxides (%)
SiO ₂	56.34
Al ₂ O ₃	23.08
Fe ₂ O ₃	6.41
CaO	9.02
MgO	1.70
K ₂ O	0.56
Na ₂ O	0.28
SO ₃	Nil
Cl	0.025
Sum	97.42
Residue on 45 µm	30 to 40
Unburnt carbon	less than 3%
Moisture	less than 1

Table 3
Density of raw materials.

Material	Density (kg/m ³)
Clay	1200
Fly Ash	770
NaOH	1350
Na ₂ SiO ₃	1460

Table 4. The amount of fly ash presents the specimen identity, e.g. FA-20 means fly ash was used by an amount of 20 wt%. In liquids, the amount of NaOH and Na₂SiO₃ was 7 wt% and 18 wt%, respectively. About 2 wt% water was also used during mixing of specimen. It can be observed from **Table 4** that the liquid amount was constant for all types of mixtures.

Specimen was prepared by mixing clay and fly ash in dry state until it attained a homogeneous color. The alkaline activator solution was used for wet mixing, NaOH and Na₂SiO₃ solution were used simultaneously. The paste was mixed with the help of electric mixer for 2–3 min. Cube molds of size 50 mm were prepared, the slurry was poured into 3 layers with simultaneous compaction in order to ensure homogenous poring and perfect surface finishing. The air voids were removed by adequate compaction through vibrating table in the laboratory. The samples, while keeping in molds, were sealed with polythene sheets and put in oven at 70 °C for 24 h. Then samples were demolded and further cured in oven at 110 °C for next 24 h. Then samples were placed at laboratory condition for testing.

2.3. Test

Physical and mechanical properties of geopolymer green bricks were investigated at macro-scale level by conducting several tests to evaluate; density, porosity, water absorption and compressive strength, in the laboratory. At micro-level, the degree of geopoly-

merization and mineralogical compositions were assessed. The detail of each test at macro and micro scale have been explained in the following sub-sections.

2.3.1. Density and porosity

Bulk density of each sample was calculated by mass to volume ratio according to BS EN 12390-7:2009 [31]. Coupled with that, true density of the samples was calculated according to ASTM D854-14 [32] by using the Eq. (1), where, m_1 is weight of pycnometer and water, m_2 is the weight of m_1 and powder sample outside the pycnometer and m_3 is weight of m_1 and powder sample inside the pycnometer. Apparent density is calculated by using the sample in pallet form, whereas; true density is calculated by using the sample in powder form. Open porosity takes open pores of the sample into account, Closed Porosity considers closed pores as well, whereas in case of total porosity all types of pores are included. The total porosity of all the specimens was calculated from the Eq. (2), whereas the close and open porosities were measured from Eq. (3) and Eq. (4), respectively.

$$\text{TrueDensity} = \text{Density of water} \cdot \left(\frac{m_2 - m_1}{m_2 - m_3} \right) \quad (1)$$

$$\text{TotalPorosity}(\%) = \left(1 - \frac{\text{BulkDensity}}{\text{TrueDensity}} \right) \cdot 100 \quad (2)$$

$$\text{ClosePorosity}(\%) = \left(1 - \frac{\text{ApparentDensity}}{\text{TrueDensity}} \right) \cdot 100 \quad (3)$$

$$\text{OpenPorosity}(\%) = \text{Totalporosity} - \text{Closeporosity} \quad (4)$$

2.3.2. Water absorption

Water absorption test was conducted according to ASTM C830 [33] to compare the water absorption of geopolymer green brick with that of normal brick and to further clarify the durability and quality of the geopolymer green brick. The samples were kept at 110 °C for 24 h and dry weight (W_1) of every sample was noted down. Then the samples were immersed in water for 24 h. After that the samples were taken out of water, wiped with cloth and wet weight (W_2) was noted down. Lastly, the water absorption was calculated by the following Eq. (5).

$$\text{WaterAbsorption}(\%) = \left(\frac{W_2 - W_1}{W_1} \right) \cdot 100 \quad (5)$$

2.3.3. Compressive strength

Compressive strength is the most important parameter to determine the mechanical strength of the material. The compressive strength was measured by using 2000 kN microcomputer digital display hydraulic universal testing machine in accordance with ASTM C 109 [34].

Table 4
Composition of eleven mixtures of geopolymer green brick.

Sample	Fly ash (wt.%)	Clay (wt.%)	Fly Ash (kg/m ³)	Clay (kg/m ³)	NaOH (kg/m ³)	Na ₂ SiO ₃ (kg/m ³)
FA-00	0.0	73.0	0.0	876.0	94.5	262.8
FA-10	7.3	65.7	56.2	788.4	94.5	262.8
FA-20	14.7	58.4	113.1	700.8	94.5	262.8
FA-30	21.9	51.1	168.6	613.2	94.5	262.8
FA-40	29.2	43.8	224.8	525.6	94.5	262.8
FA-50	36.5	36.5	281.0	438.0	94.5	262.8
FA-60	43.8	29.2	337.2	350.4	94.5	262.8
FA-70	51.1	21.9	393.4	262.8	94.5	262.8
FA-80	58.4	14.7	449.6	176.4	94.5	262.8
FA-90	65.7	7.3	505.8	87.6	94.5	262.8
FA-100	73.0	0.0	562.1	0.0	94.5	262.8

2.3.4. Fourier transform infrared spectroscopy and X-ray diffraction

Fourier transform infrared spectroscopy (FTIR) analysis of each composition was done on JASCO FTIR-4100. The FTIR spectra was scanned ($400 - 4000\text{ cm}^{-1}$) with resolution of 4 cm^{-1} and scan rate was 2 mm/sec . In order to analyze different phases during geopolymerization, X-ray Diffraction is performed by Philips PW3710 MPD Control with Cu K α radiations that were generated at 20 mA and 40 kV . The summary of materials and tests conducted in this work has been summarized in Fig. 1.

3. Results and discussion

The physical and mechanical properties of geopolymer green bricks were investigated. The summary of all experimental results is given in Table 5 along with respective standard deviations. The discussion on each property has been made separately and presented in following sub-sections. The results, for better understanding, are divided into macro-scale and micro-scale properties. Macro-scale properties comprise physical and mechanical properties, whereas; micro-scale properties cover FTIR spectroscopy and XRD analysis.

3.1. Macro-Scale properties

3.1.1. Density and porosity

In order to determine the feasibility of using a material for construction, the parameter of bulk density is of paramount importance. Lesser bulk density means more porous material while

being lighter in weight and vice versa. Bulk density and porosity are inter-related in inverse manner, therefore, if the material is porous in nature, its bulk density will be smaller. As shown in Table 3, fly ash has very low density that is why it is a lightweight material, consequently, by enhancing the amount of fly ash the resultant brick becomes lighter in nature and the density of that particular sample decreases. Fig. 2 gives a clearer picture of the trend that by increasing the amount of fly ash the bulk density decreases. According to Chinese National Standard the bulk density of the clay brick ranges from $1800\text{--}2000\text{ kg/m}^3$ (Table 6), it is clear from the Fig. 2 that the bulk density of all types of geopolymer green brick is lesser than that for conventional clay bricks ($1800\text{--}2000\text{ kg/m}^3$), which brings the conclusion that the geopolymer green brick is lighter in weight, that will ensure lightweight construction.

Bulk density of the brick is the density inclusive of open and closed pores, whereas; the apparent density is inclusive of closed pores only, and the true density is exclusive of all the pores. These densities were used to calculate open, closed and total porosity. The trends of bulk, apparent and true density are given in Fig. 2. The least value of bulk density was obviously observed for all types of geopolymer green brick specimens as compared to the apparent and true density. Whereas, the true density is maximum at all amounts of fly ash as compared to bulk and apparent density.

The open, closed and total porosity of geopolymer green bricks are given in Fig. 3, it was observed that the trend of increase in porosity with the fly ash is consistent for three types of porosities; open, close and total. It was obvious that the total porosity will be

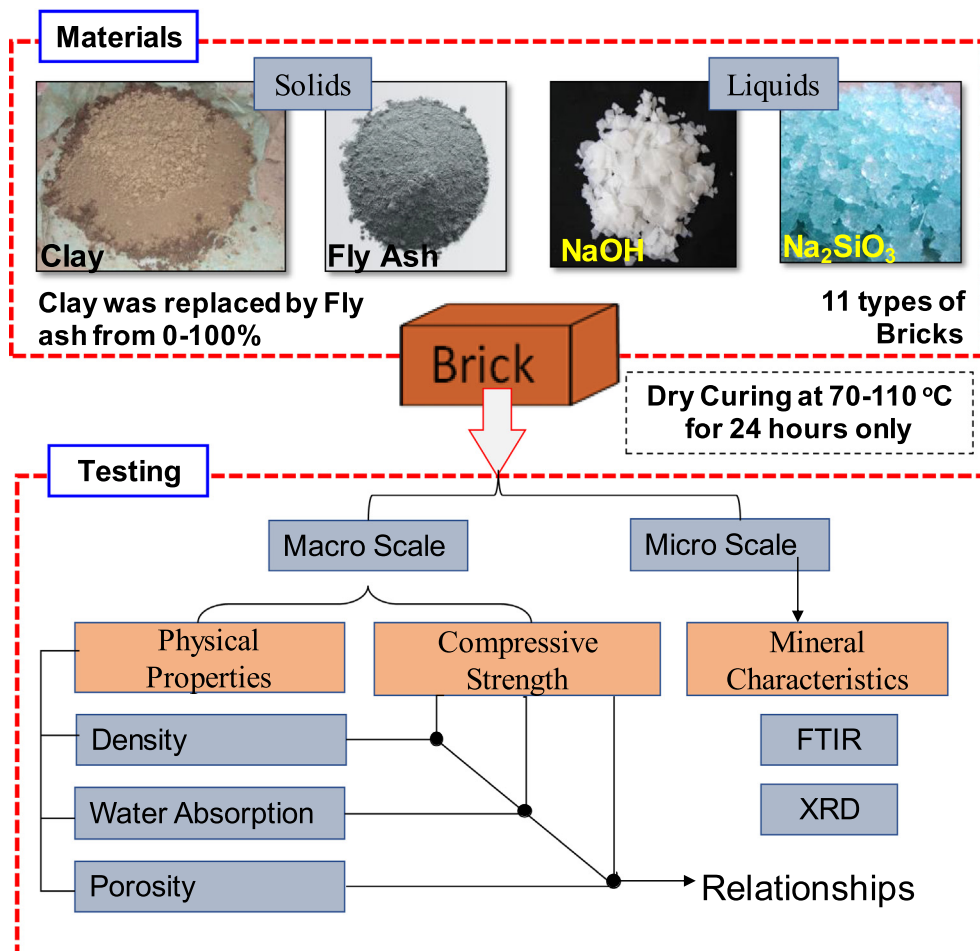


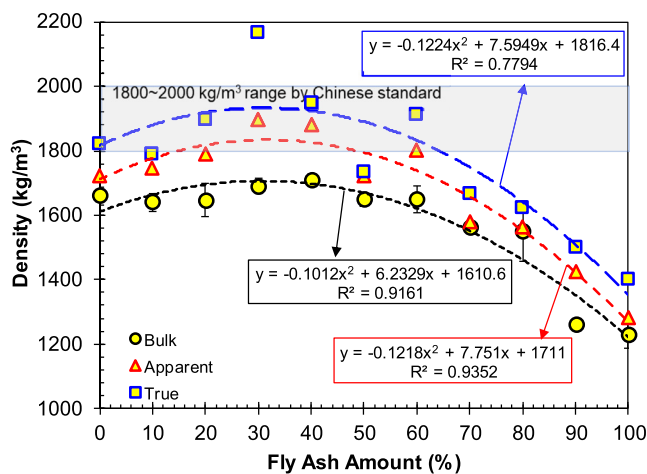
Fig. 1. Summary of the methodology of this work.

Table 5

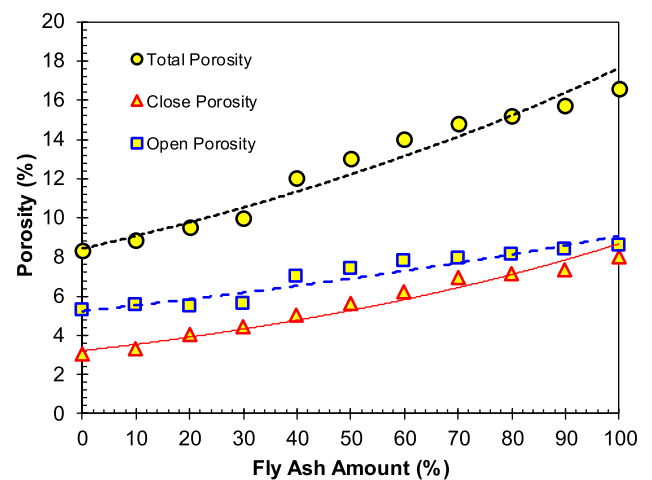
Physical and mechanical properties of geopolymers green bricks.

Specimen ID	Bulk Density (kg/m ³)	SD.	Apparent Density (kg/m ³)	True Density (kg/m ³)	Water Absorption (%)	SD.	Comp. Strength (MPa)	SD.
FA-00	1660	26.97	1721	1821	13.51	0.54	2.91	0.20
FA-10	1640	28.88	1746	1788	12.67	1.28	4.43	0.59
FA-20	1647	52.85	1789	1895	11.09	0.89	11.37	0.81
FA-30	1690	23.30	1894	2164	11.85	0.42	15.15	1.25
FA-40	1710	15.20	1878	1946	11.61	1.50	12.90	1.50
FA-50	1650	21.69	1720	1731	13.00	1.11	15.07	0.80
FA-60	1650	41.18	1800	1913	13.65	1.52	17.10	0.39
FA-70	1560	1.53	1579	1667	16.30	0.65	10.77	1.50
FA-80	1550	93.44	1563	1620	22.43	0.43	6.50	0.33
FA-90	1260	7.49	1425	1500	31.86	1.26	2.74	0.11
FA-100	1230	44.64	1280	1400	32.55	1.61	3.18	0.27

SD. = standard deviation; Comp. = compressive.

**Fig. 2.** Bulk, apparent and true density of geopolymers green bricks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

me more and close porosity will be least, which was strictly followed in all type of specimens. As, it has been said earlier, fly ash is a highly porous material while being lesser dense as compared to clay, therefore, with an increase in the amount of fly ash, porosity increases. The cenospheres being semi-crystalline in nature, get dissolved leaving behind huge amount of activated mass depending upon the relative percentage of fly ash used [35]. This activated mass gives rise to higher and higher degree of geopolymerization. Hence, higher the percentage of fly ash, higher the degree of geopolymerization. But as the cenospheres are centrally hollow, so this addition of emptiness rises with the rise of fly ash. This fly ash derived emptiness contributes to overall porosity of geopolymer which in turn decrease the density of the consequent material [36].

**Fig. 3.** Open, closed and total porosity of geopolymers green bricks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.1.2. Water absorption

Water absorption is a significant feature to assess the durability, thermal conductivity and quality of a brick. If the particles at micro-structure level are jam-packed, they would not let water to infiltrate, moreover density will be higher in that case and vice versa. The water absorption of the geopolymers green bricks is checked with reference to Indian National Standard and Chinese National Standard (Table 6). It is obvious from the results that due to fly ash being a porous material, the water absorption rate increases with an increase in percentage of fly ash. It is shown in Fig. 4 that the water absorption of geopolymers green bricks with 0–60 wt% fly ash is in the region of 1st class bricks, whereas; the water absorption of geopolymers green brick with 70% fly ash is in the region of 2nd class brick and the samples with 90–100 wt % fly ash are above 2nd class brick zone in terms of water absorp-

Table 6

Different Standard used in this study testing.

Standard	Brick Description	Bulk Density (kg/m ³)	Water Absorption (%)	Compressive Strength (MPa)
Chinese National Standard	1st Class Brick	1800–2000	15 (Max.)	15
	2nd Class Brick	1800–2000	19 (Max.)	9.8
ASTM	Severe Weathering	-----	-----	20.7
	Moderate Weathering	-----	-----	17.2
	Negligible Weathering	-----	-----	10.3
Indian Standard	1st class	-----	15 (Max.)	5–10 (load bearing)
	2nd Class	-----	20 (Max.)	3–5 (non-load bearing)
Brazilian Standard	-----	-----	-----	1.5 (Min.)

Max. = maximum; Min. = minimum.

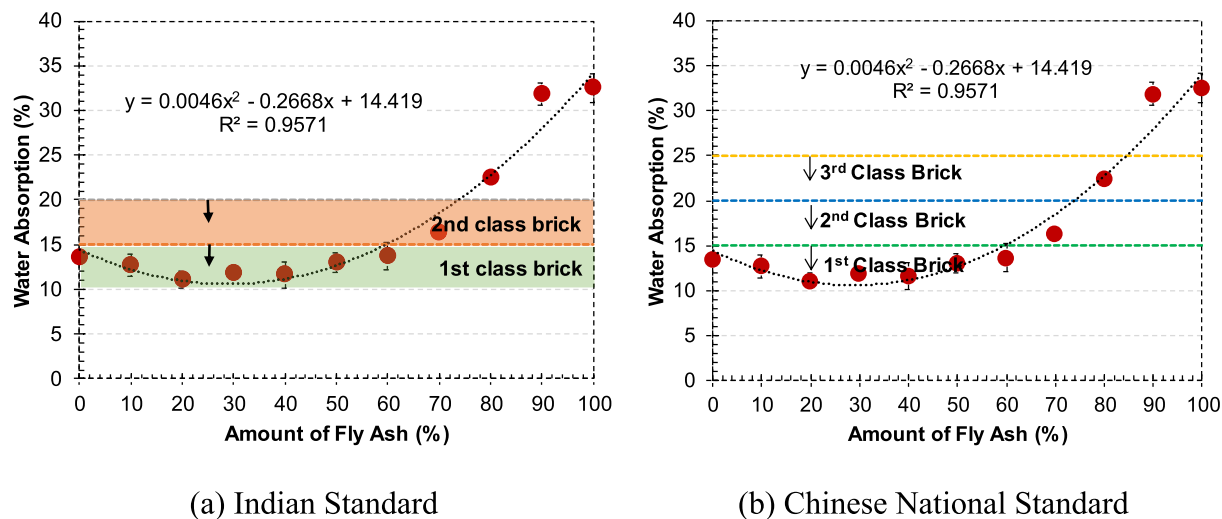


Fig. 4. Water absorption by all types of geopolymer green bricks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

tion. Coupled with that, when compared to Chinese National Standard for water absorption, the geopolymer green bricks with 0–60 wt% fly ash are in the zone of 1st class brick, the brick with 70 wt% fly ash is in the zone of 2nd class, the brick with 80 wt% in 3rd class, while the geopolymer green brick with 90–100 wt% fly ash fall above the 3rd class brick zone in terms of their water absorption percentage. The reason for the trend, as stated earlier is attributed to the fly ash being more porous and less dense in nature than clay. In another manner, it can be said that by decreasing the amount of clay, water absorption increases. When hollow shells of fly ash are activated their surfaces are dissolved (being semi-crystalline in nature) [37]. As soon as the surfaces are activated/dissolved, cenosphere leave behind hollow skeletons, which actually cause generation of porosity stepwise increasing with the increase in relative fly ash content [38]. The regression analysis was also conducted and relationship of water absorption by geopolymer green brick with the amount of fly ash was established and presented in Fig. 4, the value of coefficient of determination (R^2) was 0.9571, which present a significant influence of the amount of FA on the water absorption by geopolymer green brick.

3.1.3. Compressive strength

Compressive strength is the most important factor while determining the durability, feasibility and mechanical strength of a material, it was measured for all compositions of bricks and presented in Fig. 5. The compressive strength ranged from 2.73 MPa to 17.10 MPa (Table 5). It was observed that by increasing the amount of fly ash the compressive strength of corresponding sample was increased, but this tendency exists up to 60 wt% fly ash, afterwards upon further increasing the content of fly ash, compressive strength was decreased.

The samples with zero percent fly ash (FA-00) exhibited less strength because clay is a hydrothermal product in which silica and alumina was present in tetrahedral symmetry and in the process of geopolymerization only surface of clay is activated, moreover, in case of normal brick manufacturing 25 wt% water is added for mixing purposes and in this study only 2 wt% of water was added due to which shrinkage cracks appeared (Fig. 6) at the surface of geopolymer green brick samples with 0 and 10 wt% fly ash, furthermore, conventional clay brick was cured at 1000 °C–1500 °C, whereas; the geopolymer samples were cured at 70 °C–110 °C. The reason for this peculiar trend is that the fly ash is a better material for geopolymerization than clay, keeping the fact clay

need higher sintering temperature aside. At 80–100 wt% fly ash the samples again displayed lesser compressive strength as fly ash is lightweight, porous material as well as the cenospheres at microlevel are hollow and cracked that prohibit the material from achieving higher compressive strength. The relationship of compressive strength with the amount of fly ash was also established by applying regression analysis (polynomial), and concerned equation and R^2 are mentioned in Fig. 5.

Afterwards the compressive strength of geopolymer green bricks was tested against four standards; CNS 382 [39], ASTM C62-13a, 2013 [40], Brazilian Standard NBR 6064 (ABNT 1983a) [41] and Indian Standard (IS) 1077 (2007) [42], the summary of the range for compressive strength is mentioned in Table 6. The Indian and Chinese Standards are used as both countries have same environmental, weathering, domestic and constructional conditions as Pakistan, whereas; ASTM is a renowned universally accepted standard. Chinese National Standard divides bricks into two categories named as 1st class and 2nd class bricks, with compressive strength 14.7 MPa for 1st class and 9.8 MPa for 2nd class. In comparison with Chinese National Standard it was found that the samples with 30, 50 and 60 wt% fly ash fall above the region of 1st class brick, whereas the bricks with 20, 40 and 70% fly ash are above 2nd class brick zone and 0, 10, 80–100 wt% fly ash are below 2nd class brick region as mentioned in Fig. 5(a). The Indian Standard categorizes the compressive strength into two ranges, i.e. load-bearing range and non-load-bearing range. A keener look at geopolymer green bricks casted in this study in comparison with Indian Standard for compressive strength shows that bricks with 0, 10, 90 and 100% fly ash are in the region of non-load-bearing range, whereas; the bricks with 20–70 wt% fly ash fall above load-bearing range (Fig. 5(b)). Furthermore, when checked with reference to ASTM standard for compressive strength the geopolymer green bricks with 30–70% fly ash fall between negligible weathering and moderate weathering zone, while the geopolymer green bricks with 0–10 wt% fly ash, 80–100 wt% fly ash are below negligible weathering range (Fig. 5(c)). Finally, Brazilian Standard sets a minimum limit of 1.5 MPa, so all of the geopolymer green bricks are above this minimum standard as shown in Fig. 5(d).

3.1.4. Relationship between physical and mechanical properties

Relationship between bulk density and water absorption was established and presented in Fig. 7, the bulk density has the inverse relationship with the water absorption. It was discussed in

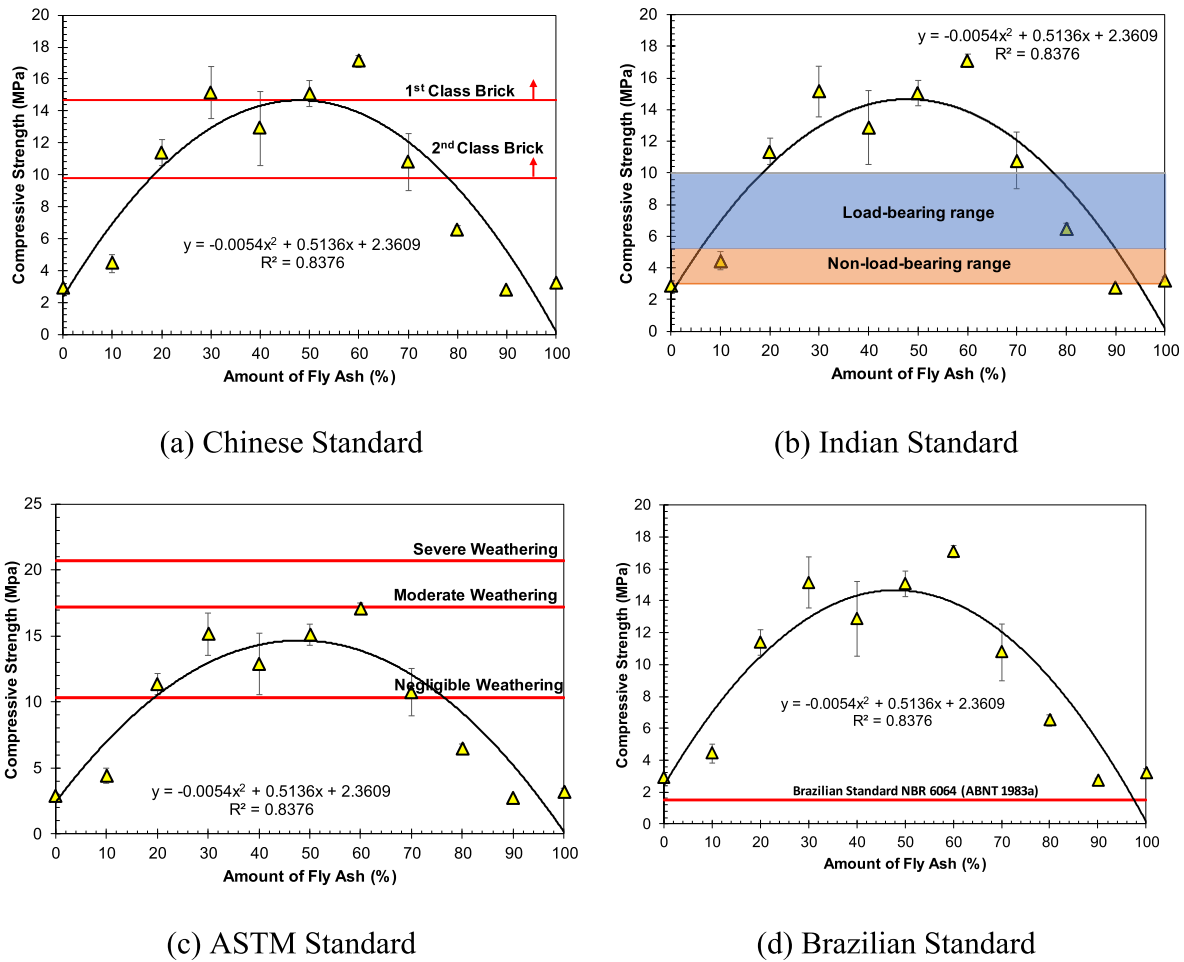


Fig. 5. Compressive Strength of all types of geopolymers green brick. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



Fig. 6. Shrinkage cracks appeared at the surface of samples (FA-00).

Section 3.1.1 and 3.1.2 that density decreases and water absorption increases with the increase in the amount of fly ash, respectively. Fig. 7 presents this relationship and both physical properties verifies the results, as it can be easily observed that up to 60 wt% of fly ash water absorption is within allowable range and low density results in light weight geopolymers green brick, as described by codes (Table 6).

Relationships were also established between physical and mechanical properties, Fig. 8(a) present the relationship between compressive strength and bulk density under the influence of amount of fly ash. It is obvious that compressive strength increases with the increase in bulk density as the dense microstructure formed. Similar trend was observed in Fig. 8(a), the less reduction at low fly ash amount was due to shrinkage cracks as explained in Fig. 6 and Section 3.1.3. Coupled with that, relationship of com-

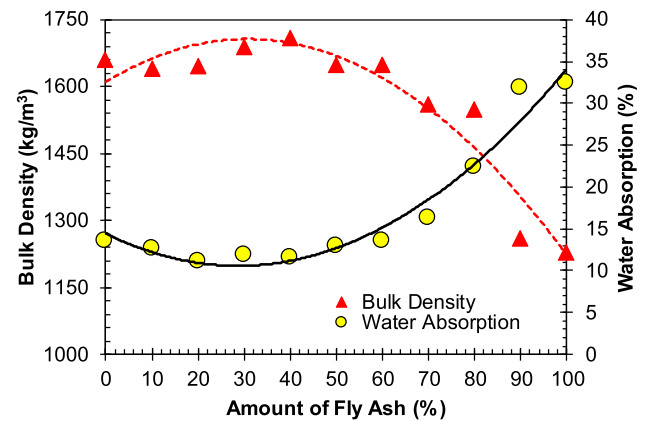


Fig. 7. Relationship between bulk density and water absorption of geopolymers green brick. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

pressive strength with water absorption is presented in Fig. 8(b), it has been noticed that with an increase in the water absorption the compressive strength decreases. To put it another way, as fly ash is hydrophilic in nature, that is the reason, with an increase in the amount of fly ash the water absorption of geopolymers green bricks increases.

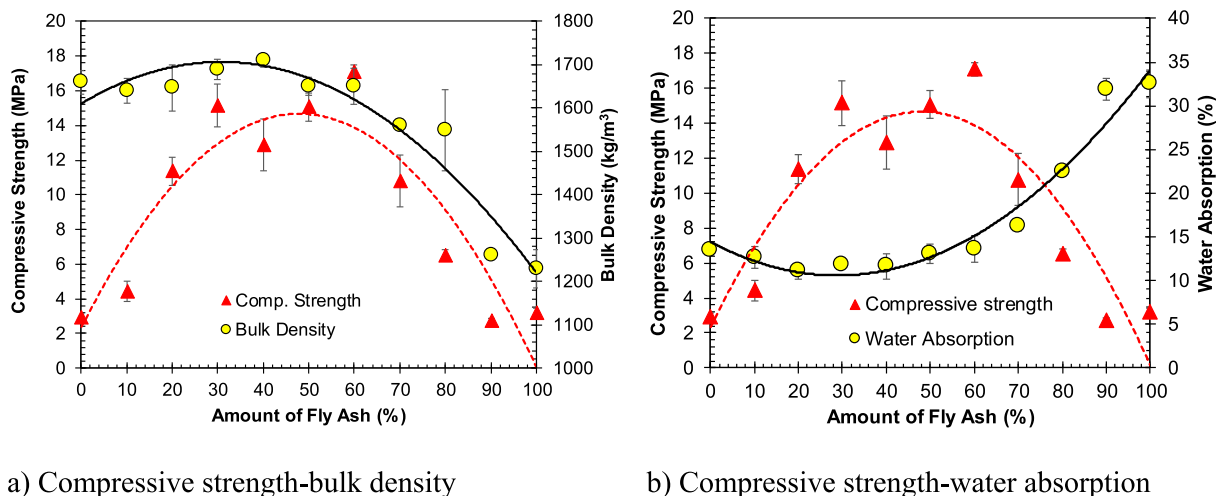


Fig. 8. Relationship between physical and mechanical properties of geopolymer green brick. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Micro-scale properties

3.2.1. FTIR analysis

FTIR shows that the region from 900 cm^{-1} to 1100 cm^{-1} is the backbone of the structure, Si-O-Al bonds. The bending at 1490 cm^{-1} shows H-O-H bonds, the presence of water [43]. Clay is a hydrothermal product formed by the weathering of silicate bearing rocks. Owing to the trait of clay being hydrothermal the alumina is already in tetrahedral therefore negligible shift in wavelength of backbone region of geopolymer green brick samples was observed (Fig. 9). In addition, the -OH bending at 1490 cm^{-1} showed the presence of water in clay. On the other hand, the backbone region of fly ash showed bending at 1100 cm^{-1} that points out the octahedral structure of alumina in fly ash. In addition to that it has been noticed that the FTIR of fly ash shows no -OH bending at 1490 cm^{-1} that concludes fly ash does not contain water and it is not hydrothermal product that is formed in the absence of water due to heating. In this perspective, when we observe the FTIR of geopolymer bricks it is obvious that the -OH bending appears with the increase in clay content. The finger-

print region of fly ash shows bending at 1100 cm^{-1} , whereas; shift towards lower wavenumber is noticed in the fingerprint region of green bricks that ensures the geopolymerization. For the FA-00 where fly ash was zero percent and 73 percent or the entire solid precursor was clay, the Si-O-T peak appeared around 990 cm^{-1} while moving towards FA-60 the peaks successively passed towards 1000 cm^{-1} . Although the difference was minor but still it was there owing to the fact that clay precursors are naturally products of hydrothermal reactions. While fly ash is produced by the fusion of aluminosilicates at higher temperatures in the furnaces, where the size of the molecules are smaller or glassy structure is dominant as indicated by the higher wavenumbers in fly ash; around 1100 cm^{-1} (Fig. 9).

Fig. 10 shows that the oxygen in raw materials is bridged oxygen. During the process of activation, the bridged oxygen changes into non-bridged oxygen. After geopolymerization, due to polycondensation the non-bridged oxygen, once again, changed to bridged oxygen [44]. Fig. 11 shows degree of geopolymerization, which was quantified by measuring the area of the peak of FTIR curves (Fig. 9) [45]. By keeping the area under the peak of fly ash curve as reference, all samples of geopolymer green brick verify the mechanism of geopolymerization. The samples FA-40, FA-50 and FA-60 shows higher degree of geopolymerization and in combination to density and porosity, they possess higher compressive strength. The reduction in compressive strength with higher amount of fly ash, although with high degree of geopolymerization, was due to more porous nature of the material as explained in Section 3.1.3.

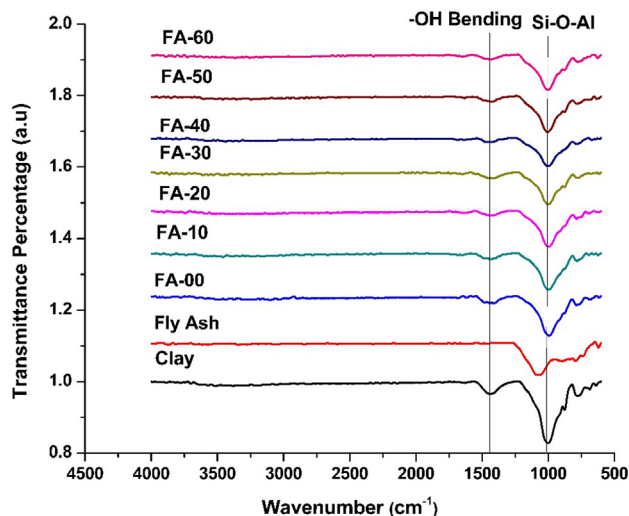


Fig. 9. FTIR of clay, fly ash and geopolymer green bricks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2.2. XRD analysis

In order to investigate different phases present in geopolymer green bricks, XRD analysis was performed. Fig. 12 shows the results of XRD, the results of XRD compliment the results of FTIR. The peak of anorthite (A) as highlighted in Fig. 12 shows the phenomenon of geopolymerization. The peak of anorthite is absent in fly ash that shows fly ash is a material produced as process of burning. On the other hand, the presence of anorthite in clay shows it is a naturally hydrothermal and the oxides of silica and alumina are in tetrahedral symmetry [45]. The presence of calcite (C) is confirmed in the XRD analysis [46]. In process of geopolymerization of clay, only surface is activated as silica and alumina is already in tetrahedral symmetry. The surface activation in clay changes the crystallinity of material. The more prominent curve of anor-

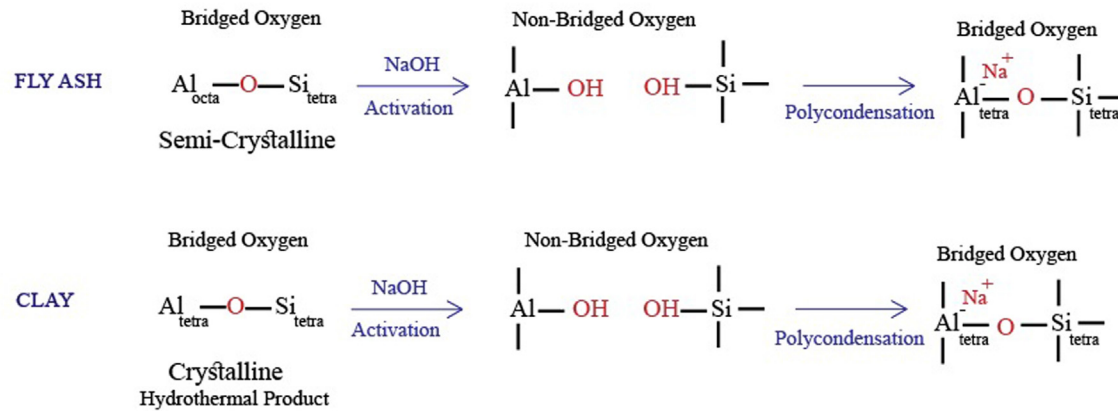


Fig. 10. Mechanism of geopolymerization in fly ash and clay.

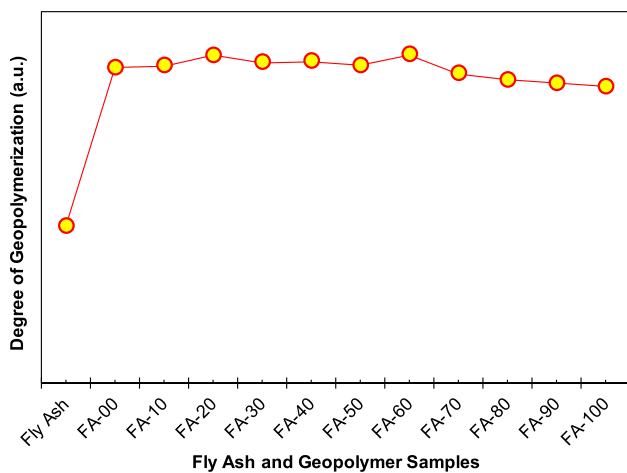


Fig. 11. Degree of geopolymerization for fly ash and geopolymer green bricks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

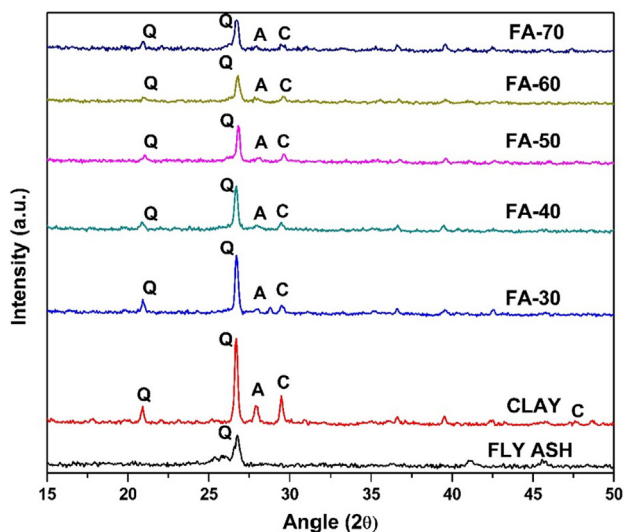


Fig. 12. XRD of clay, fly ash and resultant geopolymer samples. (Note: A = Anorthite, Q = Quartz and C = Calcite).

thite in FA-40, FA-50 and FA-60 marked higher degree of geopolymerization consequently higher compressive strength.

4. Conclusions

In this research work, bricks were prepared by replacing the conventional raw material (clay) by the fly ash, and strength was gained by the geopolymerization. Eleven mix proportions were designed to assess the physical and mechanical properties of geopolymer green bricks at macro-scale. The chemical structure and minerals in the geopolymer bricks were assessed through FTIR spectroscopy and XRD analysis. Based on the experimental study carried out on geopolymer green bricks, the following extrapolations are drawn:

Bulk, apparent and true densities reduce with the addition of fly ash, the maximum bulk density was less than 1700 kg/m^3 , which is obviously less than range described in Chinese standard ($1800\text{--}2000 \text{ kg/m}^3$), consequently, will lead to light weight sustainable material.

Porosity and water absorption increase with the fly ash addition, up to 60 wt% addition of fly ash, the water absorption by geopolymer green brick was less than 15 wt% and can be categorized as 1st class brick according to Indian and Chinese standards.

Compressive strength was assessed for all types of geopolymer green bricks and compared with four codes. The distinct compressive strength is exhibited by the bricks having fly ash amount from 30 to 60 wt% and can be used for load-bearing construction.

Relationships among physical and mechanical properties with the addition of fly ash were established and results are consistent with each other. Compressive strength was obviously increased with increasing bulk density and reduced with increasing water absorption.

FTIR analysis confirmed the geopolymerization, therefore wavenumber moved from higher to lower value due to polycondensation, which attributed to increase in compressive strength. The presence of Anorthite in XRD of geopolymer samples also confirms the phenomenon of geopolymerization. In addition, the results of XRD compliment the results of FTIR.

Henceforth, it can be easily concluded that geopolymer green bricks are lightweight and eco-friendly in nature. It has the almost similar compressive strength parameter and reduced water absorption factor when compared to conventional clay bricks. Further work is still needed to formulate geopolymer green brick by pressing instead of molding.

CRedit authorship contribution statement

Sahar Iftikhar: Methodology, Investigation, Formal analysis. **Khuram Rashid:** Conceptualization, Supervision. **Ehsan Ul Haq:** Supervision. **Idrees Zafar:** . **Fahad K. Alqahtani:** . **M. Iqbal Khan:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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